

Problem 1 – Question 5

Gate construction

5(a): SWAP from CNOT gates

Use the chronological sequence

$$\boxed{\text{CNOT}_{1 \rightarrow 2}, \quad \text{CNOT}_{2 \rightarrow 1}, \quad \text{CNOT}_{1 \rightarrow 2} .}$$

On a computational basis state $|x, y\rangle$, the bit values evolve as

$$(x, y) \mapsto (x, x \oplus y) \mapsto (y, x \oplus y) \mapsto (y, x).$$

Linearity gives the same result for arbitrary states $|\psi\rangle|\phi\rangle$. Three CNOTs are necessary and sufficient if CNOT is the only available two-qubit gate.

5(b): Two distance-two CNOTs on a four-qubit line

Assume the desired directions are $q_0 \rightarrow q_2$ and $q_1 \rightarrow q_3$. The following nearest-neighbor schedule uses six CNOTs:

layer	gates
1	$\text{CNOT}_{0 \rightarrow 1}$
2	$\text{CNOT}_{1 \rightarrow 2}$
3	$\text{CNOT}_{0 \rightarrow 1}$ and $\text{CNOT}_{2 \rightarrow 3}$
4	$\text{CNOT}_{1 \rightarrow 2}$
5	$\text{CNOT}_{2 \rightarrow 3}$.

The two gates in layer 3 act on disjoint pairs and run in parallel. Tracking the four classical basis bits gives

$$(x_0, x_1, x_2, x_3) \mapsto (x_0, x_1, x_2 \oplus x_0, x_3 \oplus x_1),$$

so the induced unitary is exactly the product of the two requested CNOTs. The count is 6 and the depth is 5. A SWAP-routing construction would use 8 CNOTs, so preserving the temporary parity information saves two gates.

5(c): R_{ZZ} from CNOT and R_Z

Because

$$\text{CNOT}_{1 \rightarrow 2} (\mathbb{I} \otimes Z) \text{CNOT}_{1 \rightarrow 2} = Z \otimes Z,$$

conjugating an R_Z rotation gives

$$\boxed{R_{ZZ}(\theta) = \text{CNOT}_{1 \rightarrow 2} (\mathbb{I} \otimes R_Z(\theta)) \text{CNOT}_{1 \rightarrow 2} .}$$

Chronologically: CNOT $1 \rightarrow 2$, $R_Z(\theta)$ on qubit 2, then the same CNOT again. The circuit uses two CNOTs and one R_Z .

5(d): Controlled phase from R_{ZZ}

The projector onto $|11\rangle$ is

$$|11\rangle\langle 11| = \frac{1}{4}(\mathbb{I} - Z_1)(\mathbb{I} - Z_2) = \frac{1}{4}(\mathbb{I} - Z_1 - Z_2 + Z_1 Z_2).$$

Since all four terms commute,

$$\begin{aligned} \text{CP}(\theta) &= \exp(i\theta|11\rangle\langle 11|) \\ &= e^{i\theta/4} R_{Z,1} \left(\frac{\theta}{2} \right) R_{Z,2} \left(\frac{\theta}{2} \right) R_{ZZ} \left(-\frac{\theta}{2} \right). \end{aligned}$$

Hence, up to the physically irrelevant global phase $e^{i\theta/4}$,

$$\boxed{\text{CP}(\theta) \equiv R_{Z,1} \left(\frac{\theta}{2} \right) R_{Z,2} \left(\frac{\theta}{2} \right) R_{ZZ} \left(-\frac{\theta}{2} \right).}$$

The resulting diagonal entries are $1, 1, 1, e^{i\theta}$, after the global phase is restored.

5(e): Exponential of an arbitrary Pauli string

Let

$$P = P_1 \otimes \cdots \otimes P_n, \quad P_j \in \{\mathbb{I}, X, Y, Z\},$$

and let $S = \{j : P_j \neq \mathbb{I}\}$ have size w . For each $j \in S$, apply a basis change B_j satisfying $B_j P_j B_j^\dagger = Z$:

$$B_j = \begin{cases} H, & P_j = X, \\ HS^\dagger, & P_j = Y, \\ \mathbb{I}, & P_j = Z. \end{cases}$$

For Y , the chronological order is $S^\dagger$ and then H . After these changes, P has become a product of Z 's.

Choose one qubit $t \in S$ as a parity target. Apply CNOTs from every other $j \in S \setminus \{t\}$ to t , apply $R_Z(2\theta)$ to t , undo the CNOTs in reverse order, and finally undo all B_j . The middle rotation is

$$R_Z(2\theta) = e^{-i\theta Z_t},$$

and parity conjugation replaces Z_t by $\prod_{j \in S} Z_j$. Thus the complete circuit implements

$$\boxed{e^{-i\theta P}}.$$

It uses $2(w-1)$ CNOTs, one R_Z , and at most $2w$ one-qubit basis-change gates. A balanced parity tree reduces the CNOT depth from $2(w-1)$ to $2\lceil \log_2 w \rceil$ when connectivity permits.

5(f): The specified two-qubit matrix

Use computational-basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, and define

$$R_Y(\vartheta) = \begin{pmatrix} \cos(\vartheta/2) & -\sin(\vartheta/2) \\ \sin(\vartheta/2) & \cos(\vartheta/2) \end{pmatrix}.$$

The requested real matrix U has the exact factorization

$$U = \text{CZ} \left[\mathbb{I} \otimes R_Y \left(-\frac{13\pi}{12} \right) \right] \text{CZ} \\ \times \left[R_Y \left(\frac{3\pi}{4} \right) \otimes R_Y \left(-\frac{5\pi}{12} \right) \right] \text{CZ} \left[R_Y \left(\frac{3\pi}{4} \right) \otimes \mathbb{I} \right].$$

Thus the chronological pulse schedule is

$$\begin{array}{l|l} 1 & R_Y(3\pi/4) \text{ on } q_0 \\ 2 & \text{CZ} \\ 3 & R_Y(3\pi/4) \text{ on } q_0 \quad \text{and} \quad R_Y(-5\pi/12) \text{ on } q_1 \\ 4 & \text{CZ} \\ 5 & R_Y(-13\pi/12) \text{ on } q_1 \\ 6 & \text{CZ}. \end{array}$$

Direct multiplication yields

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} & 0 \\ \frac{1}{2\sqrt{2}} & 0 & -\frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2\sqrt{2}} & 0 & -\frac{\sqrt{3}}{2\sqrt{2}} & -\frac{1}{2} \end{pmatrix}.$$

The two simultaneous rotations in layer 3 give a total circuit depth of 6, with three CZ gates and four R_Y pulses.

Determinant note. As printed, the matrix is real orthogonal with determinant -1 , not $+1$. Therefore it is not literally in $SU(4)$. Multiplying it by the global phase $e^{i\pi/4}$ gives determinant 1, because $\det(e^{i\pi/4}U) = e^{i\pi} \det U = 1$. This global phase has no observable effect, so the displayed circuit implements the intended two-qubit operation.